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Cryptology Theory 6

(1) Caesar wants to arrange a secret ~~meaning~~ meeting with Mark Antony, either at the Timber River or at the Colosseum. He sends the ciphertext EVIRE. However, Antony does not know the key, so he tries all possible keys.

a) What are the meaningful plaintexts Antony obtains?

b) Where should Antony meet Caesar?

c) What interesting point about a ciphertext-only attack do you learn from this exercise?

→ Ciphertext values

E	V	I	R	E
4	21	8	17	4

Decryption rule for key k:

$P = (C - k) \bmod 26$

Two worked trials

$k = 4$

$E = (4-4) \bmod 26 = 0 \rightarrow A$

$V = (21-4) \bmod 26 = 17 \rightarrow R$

$I = (8-4) \bmod 26 = 4 \rightarrow E$

$R = (17-4) \bmod 26 = 13 \rightarrow N$

$E = (4-4) \bmod 26 = 0 \rightarrow A$

Result: ARENA

$k = 13$

$E = (4-13) \bmod 26 = -9 \bmod 26 = 17 \rightarrow R$

$V = (21-13) \bmod 26 = 8 \rightarrow I$

$I = (8-13) \bmod 26 = -5 \bmod 26 = 21 \rightarrow V$

$R = (17-13) \bmod 26 = 4 \rightarrow E$

$E = (4-13) \bmod 26 = -9 \bmod 26 = 17 \rightarrow R$

Result: River

All 26 keys:

k	Plaintext	k	Plaintext
0	EVTRE	13	RIVER
1	DUHQD	14	QHUDQ
2	CTGPC	15	PGTCP
3	BSFOB	16	OFSBO
4	ARENA	17	NERAN
5	ZQDMZ	18	MDQ2M
6	YPCLY	19	LCPYL
7	XOBKX	20	KBOXK
8	WNAJW	21	JANWJ
9	VMZIV	22	IZMVI
10	ULYHU	23	HYLUH
11	TKXGT	24	GXKTG
12	SJWFS	25	FWJSF

(a) Meaningful Plaintext:
ARENA (k=4) and RIVER (k=13). The other 24 keys give gibberish

(b) Where to meet.
RIVER = TIBER River,
ARENA = Colosseum.
Both are meaningful, so Antony cannot decide from the Ciphertext alone. He must confirm with Caesar by another channel

(c) Interesting point: A Ciphertext-only attack can give more than one meaningful plaintext when the ciphertext is short. A longer ciphertext is needed for unique answer

(2) Consider an Affine Cipher operating mod 26. In a chosen-plaintext attack, the plaintext HAHAHA produces the Ciphertext NONONO

a) Convert the plaintext & Ciphertext letters into their numerical value
b) Form the required equations mod 26
c) Determine the values of a and b.
d) Write the encryption function $E(x) = (ax+b) \bmod 26$

a)

Letter	H	A	H	A	H	A
Plaintext(x)	7	0	7	0	7	0
Letter	N	O	N	O	N	O
Ciphertext(y)	13	14	13	14	13	14

The pattern repeats, so only two distinct pairs matters: $7 \to 13$ & $0 \to 14$

(b) Equations $y = ax + b$

$7a + b \equiv 13 \bmod 26$ — (1)

$0a + b \equiv 14 \bmod 26$ — (2)

(c) Solve:

From (2): $b = 14$

Put in (1): $7a + 14 \equiv 13 \to 7a \equiv -1 \equiv 25 \pmod{26}$

$7^{-1} \bmod 26 = 15$ (since $7 \times 15 = 105 = 4 \times 26 + 1$)

$a \equiv 15 \times 25 = 375 = 14 \times 26 + 11 \to a = 11$

Check: $\gcd(11, 26) = 1$ ✓ $11 \times 7 + 14 = 91 = 3 \times 26 + 13 \equiv 13$

d) Encryption function

$E(x) = (11x + 14) \bmod 26$

(3) The Ciphertext CRWW2 was encrypted using an affine Cipher module 26. The attacker knows that the plaintext begins with HA.

(a) Use the known plaintext - ciphertext pair to determine a and b.

(b) Write the encryption function

(c) Find the multiplicative inverse of module 26 and write the decryptive function.

(d) Decrypt the completed Ciphertext.

→ a) Find a and b.

$H = 7 \to C = 2$ $\quad A = 0 \to R = 17$

$7a + b \equiv 2 \quad -(1)$ $\quad 0a + b \equiv 17 \quad -(2)$

From (2): $b = 17$

$7a \equiv 2 - 17 \equiv -15 \equiv 11 \pmod{26}$

$a \equiv 15 \times 11 \equiv 165 = 6 \times 26 + 9 \Rightarrow a = 9$

Check: $\gcd(9, 26) = 1$ ✓ $\quad 9 \times 7 + 17 = 80 = 3 \times 26 + 2 \equiv 2$ ✓

(b) Encryption Function:

$E(x) = (9x + 17) \bmod 26$

(c) Inverse of a, decryption function

Euclid: $26 = 2 \times 9 + 8$, $\quad 9 = 1 \times 8 + 1 \to 1 = 9 - 8 = 9 - (26 - 2 \times 9)$

$= 3 \times 9 - 26$

So $9 \times 3 = 27 \equiv 1 \to 9^{-1} = 3$

$D(y) = 3(y - 17) \bmod 26 = 2(3y - 51) \bmod 26 =$

$(3y + 1) \bmod 26 \quad (-51 + 52 = 1)$

d) Decrypt CRWW2

Cipher	y	$3(y-17) \bmod 26$	x	Plain
C	2	$3(-15) = -45 \equiv 7$	7	H
R	17	$3(0) = 0$	0	A
W	22	$3(5) = 15$	15	P
W	22	$3(5) = 15$	15	P
2	25	$3(8) = 24$	24	Y

Plaintext = HAPPY

④ Alice uses a One-Time pad to encrypt the plaintext RIVER with the key NQXTP.
a) Convert the plaintext and key into numerical values
b) Encrypt the message using $C_i = (P_i + K_i) \bmod 26$
c) Decrypt the ~~message~~ resulting ciphertext using $P_i = (C_i + K_i) \bmod 26$
d) Explain why the key must be random, equal in length to the plaintext and used only once.

→ a) Numerical Value:

	R	I	V	E	R
P	17	8	21	4	17
	N	Q	X	T	P
K	13	16	23	19	15

b) Encryption: $C = (P + K) \bmod 26$

$17 + 13 = 30 - 26 = 4 \rightarrow E$
$8 + 16 = 24 \rightarrow Y$
$21 + 23 = 44 - 26 = 18 \rightarrow S$
$4 + 19 = 23 \rightarrow X$
$17 + 15 = 32 - 26 = 6 \rightarrow G$
Ciphertext = EYSXG

c) Decryption: $P = (C - K) \bmod 26$

$4 - 13 = -9 + 26 = 17 \rightarrow R$
$24 - 16 = 8 \rightarrow I$
$18 - 23 = -5 + 26 = 21 \rightarrow V$
$23 - 19 = 4 \rightarrow E$
$6 - 15 = -9 + 26 \Rightarrow 17 \rightarrow R$
Plaintext = RIVER

(d) Random: Every plaintext is equally likely, so the ciphertext reveals nothing (perfect secrecy)

Same length: A shorter key must repeat, and repeating keys can be broken by frequency analysis.

Used Once: Reusing a key gives $C_1 - C_2 = P_1 - P_2$ so the key cancels and leaks information about both messages.